Some Cars from the gtcars Dataset

Five Cars are shown here

Everything but the cost				
Make and Model		Performance		
mfr	model	hp	trq	msrp
Ford	GT	647	550	$447,000.00
Ferrari	458 Speciale	597	398	$291,744.00
Ferrari	458 Spider	562	398	$263,553.00
Ferrari	458 Italia	562	398	$233,509.00
Ferrari	488 GTB	661	561	$245,400.00

Cars are all 2015 models.
Horsepower and Torque values are estimates.

```
import polars as pl
from great_tables import GT, md, html
from great_tables.data import islands

islands_mini = pl.from_pandas(islands).sort("size", descending=True).head(10)

print(
    GT(islands_mini)
    .tab_header(title="Large Landmasses of the World", subtitle="The top ten largest are pres
    # .tab_stub(rowname_col="name")
    .tab_source_note(source_note="Source: The World Almanac and Book of Facts, 1975, page 406
    .tab_source_note(
        # source_note=md("Reference: McNeil, D. R. (1977) *Interactive Data Analysis*. Wiley
        source_note=html("Reference: McNeil, D. R. (1977) *Interactive Data Analysis*. Wiley
    )
    .tab_stubhead(label="landmass")
    .fmt_image(columns="size")
    .as_latex()
)
```

```
\begin{table}
\caption*{
{\large Large Landmasses of the World} \\
{\small The top ten largest are presented}
}
```

Large Landmasses of the World

The top ten largest are presented

name	size
Africa	11,506
Antarctica	5,500
Asia	16,988
Australia	2,968
Axel Heiberg	16
Baffin	184
Banks	23
Borneo	280
Britain	84
Celebes	73

Source: The World Almanac and Book of Facts, 1975, page 406.
Reference: McNeil, D. R. (1977) *Interactive Data Analysis*. Wiley.

New York Air Quality Measurements

Daily measurements in New York City (May 1-10, 1973)

Ozone	Solar_R	Wind	Temp	Month	Day	Year
41.0	190.0	7.4	67	5	1	1973
36.0	118.0	8.0	72	5	2	1973
12.0	149.0	12.6	74	5	3	1973
18.0	313.0	11.5	62	5	4	1973
nan	nan	14.3	56	5	5	1973
28.0	nan	14.9	66	5	6	1973
23.0	299.0	8.6	65	5	7	1973
19.0	99.0	13.8	59	5	8	1973
8.0	19.0	20.1	61	5	9	1973
nan	194.0	8.6	69	5	10	1973

Physical Constants Having a Molar Basis

name	value
Molar Planck Constant	3.990×10^{-10}
Electron Molar Mass	5.486×10^{-7}
Molar Volume of Silicon	1.206×10^{-5}
Muon Molar Mass	1.134×10^{-4}
Molar Mass Constant	1.000×10^{-3}
Proton Molar Mass	1.007×10^{-3}
Neutron Molar Mass	1.009×10^{-3}
Tau Molar Mass	1.908×10^{-3}
Deuteron Molar Mass	2.014×10^{-3}
Helion Molar Mass	3.015×10^{-3}
Triton Molar Mass	3.016×10^{-3}
Alpha Particle Molar Mass	4.002×10^{-3}
Molar Mass of Carbon-12	1.200×10^{-2}
Molar Volume of Ideal Gas (273.15 K, 101.325 kpa)	2.241×10^{-2}
Molar Volume of Ideal Gas (273.15 K, 100 kpa)	2.271×10^{-2}
Molar Gas Constant	8.314

num	date	time	currency
111 B	Thursday, January 15, 2015	[13:35]	49.95
2.2 KiB	Sunday, February 15, 2015	[14:40]	17.95
32.5 KiB	Sunday, March 15, 2015	[15:45]	_$1.39_
434 KiB	Wednesday, April 15, 2015	[16:50]	_$65,100.00_
5.3 MiB	Friday, May 15, 2015	[17:55]	_$1,325.81_

```
\fontsize{12.0pt}{14.4pt}\selectfont

\begin{tabular*}{\linewidth}{@{\extracolsep{\fill}}lr}
\toprule
name & size \\
\midrule\addlinespace[2.5pt]
Asia & 16988 \\
Africa & 11506 \\
North America & 9390 \\
South America & 6795 \\
Antarctica & 5500 \\
Europe & 3745 \\
Australia & 2968 \\
Greenland & 840 \\
New Guinea & 306 \\
Borneo & 280 \\
\bottomrule
\end{tabular*}
\begin{minipage}{\linewidth}
Source: The World Almanac and Book of Facts, 1975, page 406.\\
Reference: McNeil, D. R. (1977) *Interactive Data Analysis*. Wiley.\$\\
\end{minipage}
\end{table}
```

```
/opt/hostedtoolcache/Python/3.10.20/x64/lib/python3.10/site-packages/great_tables/_formats.py
  warn("fmt_image() is not currently implemented in LaTeX output.")
```

```python
from great_tables import GT, html
from great_tables.data import airquality

airquality_mini = airquality.head(10).assign(Year=1973)

print(
    GT(airquality_mini)
    .tab_header(
        title="New York Air Quality Measurements",
        subtitle="Daily measurements in New York City (May 1-10, 1973)",
    )
    .tab_spanner(label="Time", columns=["Year", "Month", "Day"])
    .tab_spanner(label="Measurement", columns=["Ozone", "Solar_R", "Wind", "Temp"])
    .cols_move_to_start(columns=["Year", "Month", "Day"])
```

```
    .cols_label(
        Ozone=html("Ozone,<br>ppbV"),
        Solar_R=html("Solar R.,<br>cal/m<sup>2</sup>"),
        Wind=html("Wind,<br>mph"),
        Temp=html("Temp,<br>&deg;F"),
    )
    .as_latex()
)
```

```
\begin{table}
\caption*{
{\large New York Air Quality Measurements} \\
{\small Daily measurements in New York City (May 1-10, 1973)}
}

\fontsize{12.0pt}{14.4pt}\selectfont

\begin{tabular*}{\linewidth}{@{\extracolsep{\fill}}rrrrrrr}
\toprule
\multicolumn{3}{c}{Time} & \multicolumn{4}{c}{Measurement} \\
\cmidrule(lr){1-3} \cmidrule(lr){4-7}
Year & Month & Day & Ozone,ppbV & Solar R.,cal/m2 & Wind,mph & Temp,\&deg;F \\
\midrule\addlinespace[2.5pt]
1973 & 5 & 1 & 41.0 & 190.0 & 7.4 & 67 \\
1973 & 5 & 2 & 36.0 & 118.0 & 8.0 & 72 \\
1973 & 5 & 3 & 12.0 & 149.0 & 12.6 & 74 \\
1973 & 5 & 4 & 18.0 & 313.0 & 11.5 & 62 \\
1973 & 5 & 5 & nan & nan & 14.3 & 56 \\
1973 & 5 & 6 & 28.0 & nan & 14.9 & 66 \\
1973 & 5 & 7 & 23.0 & 299.0 & 8.6 & 65 \\
1973 & 5 & 8 & 19.0 & 99.0 & 13.8 & 59 \\
1973 & 5 & 9 & 8.0 & 19.0 & 20.1 & 61 \\
1973 & 5 & 10 & nan & 194.0 & 8.6 & 69 \\
\bottomrule
\end{tabular*}

\end{table}
```

```
from great_tables import GT
from great_tables.data import countrypops
import polars as pl
import polars.selectors as cs
```

```
# Get vectors of 2-letter country codes for each region of Oceania
oceania = {
    "Australasia": ["AU", "NZ"],
    "Melanesia": ["NC", "PG", "SB", "VU"],
    "Micronesia": ["FM", "GU", "KI", "MH", "MP", "NR", "PW"],
    "Polynesia": ["PF", "WS", "TO", "TV"],
}

# Create a dictionary mapping country to region (e.g. AU -> Australasia)
country_to_region = {
    country: region for region, countries in oceania.items() for country in countries
}

wide_pops = (
    pl.from_pandas(countrypops)
    .filter(
        pl.col("country_code_2").is_in(list(country_to_region))
        & pl.col("year").is_in([2000, 2010, 2020])
    )
    .with_columns(pl.col("country_code_2").replace(country_to_region).alias("region"))
    .pivot(index=["country_name", "region"], on="year", values="population")
    .sort("2020", descending=True)
)

print(
    GT(wide_pops)
    .tab_header(title="Populations of Oceania's Countries in 2000, 2010, and 2020")
    .tab_spanner(label="Total Population", columns=cs.all())
    #.tab_stub(rowname_col="country_name", groupname_col="region")
    .fmt_integer() # example fails because of this method
    .as_latex()
)
```

towny example

```
from great_tables import GT, html
from great_tables.data import sza
import polars as pl
import polars.selectors as cs

sza_pivot = (
```

```python
    pl.from_pandas(sza)
    .filter((pl.col("latitude") == "20") & (pl.col("tst") <= "1200"))
    .select(pl.col("*").exclude("latitude"))
    .drop_nulls()
    .pivot(values="sza", index="month", on="tst", sort_columns=True)
)

print(
    GT(
        sza_pivot,
        #rowname_col="month"
    )
    .data_color(
        domain=[90, 0],
        palette=["rebeccapurple", "white", "orange"],
        na_color="white",
    )
    .tab_header(
        title="Solar Zenith Angles from 05:30 to 12:00",
        subtitle=html("Average monthly values at latitude of 20&deg;N."),
    )
    .sub_missing(missing_text="")
    .as_latex()
)
```

```latex
\begin{table}
\caption*{
{\large Solar Zenith Angles from 05:30 to 12:00} \\
{\small Average monthly values at latitude of 20\&deg;N.}
}

\fontsize{12.0pt}{14.4pt}\selectfont

\begin{tabular*}{\linewidth}{@{\extracolsep{\fill}}lrrrrrrrrrrrrrr}
\toprule
month & 0530 & 0600 & 0630 & 0700 & 0730 & 0800 & 0830 & 0900 & 0930 & 1000 & 1030 & 1100 &
\midrule\addlinespace[2.5pt]
jan & None & None & None & 84.9 & 78.7 & 72.7 & 66.1 & 61.5 & 56.5 & 52.1 & 48.3 & 45.5 & 43
feb & None & None & 88.9 & 82.5 & 75.8 & 69.6 & 63.3 & 57.7 & 52.2 & 47.4 & 43.1 & 40.0 & 37
mar & None & None & 85.7 & 78.8 & 72.0 & 65.2 & 58.6 & 52.3 & 46.2 & 40.5 & 35.5 & 31.4 & 28
apr & None & 88.5 & 81.5 & 74.4 & 67.4 & 60.3 & 53.4 & 46.5 & 39.7 & 33.2 & 26.9 & 21.3 & 17
may & None & 85.0 & 78.2 & 71.2 & 64.3 & 57.2 & 50.2 & 43.2 & 36.1 & 29.1 & 26.1 & 15.2 & 8.
```

```
jun & 89.2 & 82.7 & 76.0 & 69.3 & 62.5 & 55.7 & 48.8 & 41.9 & 35.0 & 28.1 & 21.1 & 14.2 & 7.3
jul & 88.8 & 82.3 & 75.7 & 69.1 & 62.3 & 55.5 & 48.7 & 41.8 & 35.0 & 28.1 & 21.2 & 14.3 & 7.7
aug & None & 83.8 & 77.1 & 70.2 & 63.3 & 56.4 & 49.4 & 42.4 & 35.4 & 28.3 & 21.3 & 14.3 & 7.3
sep & None & 87.2 & 80.2 & 73.2 & 66.1 & 59.1 & 52.1 & 45.1 & 38.1 & 31.3 & 24.7 & 18.6 & 13
oct & None & None & 84.1 & 77.1 & 70.2 & 63.3 & 56.5 & 49.9 & 43.5 & 37.5 & 32.0 & 27.4 & 24
nov & None & None & 87.8 & 81.3 & 74.5 & 68.3 & 61.8 & 56.0 & 50.2 & 45.3 & 40.7 & 37.4 & 35
dec & None & None & None & 84.3 & 78.0 & 71.8 & 66.1 & 60.5 & 55.6 & 50.9 & 47.2 & 44.2 & 42
\bottomrule
\end{tabular*}

\end{table}


/opt/hostedtoolcache/Python/3.10.20/x64/lib/python3.10/site-packages/great_tables/_utils_rend
  warnings.warn(msg)


\begin{table}
\caption*{
{\large 2023 Mean **Carbon Intensity** (gCO2eq/kWh) and **Power Consumption** Breakdown (\%)
}

\fontsize{12.0pt}{14.4pt}\selectfont

\begin{tabular*}{\linewidth}{@{\extracolsep{\fill}}lcccccccccccc}
\toprule
Zone & CO2 Intensity & Hydro & Nuclear & Wind & Solar & Geothermal & Biomass & Gas & Coal & O
\midrule\addlinespace[2.5pt]
Sweden & 26 & 39.1\% & 26.8\% & 27.7\% & 0.1\% & 0.0\% & 0.4\% & 0.4\% & 0.8\% & 0.0\% & 4.6
Iceland & 28 & 69.4\% & 0.0\% & 0.0\% & 0.0\% & 30.6\% & 0.0\% & 0.0\% & 0.0\% & 0.0\% & 0.0
Quebec & 35 & 90.1\% & 2.1\% & 4.4\% & 0.0\% & 0.0\% & 1.9\% & 1.4\% & 0.0\% & 0.0\% & 0.0\%
France & 46 & 12.3\% & 65.4\% & 10.3\% & 1.8\% & 0.0\% & 1.0\% & 7.1\% & 0.3\% & 0.3\% & 0.1
Ontario & 104 & 23.3\% & 49.4\% & 8.7\% & 0.1\% & 0.0\% & 0.2\% & 18.1\% & 0.0\% & 0.0\% & 0
New Zealand & 106 & 60.5\% & 0.0\% & 7.7\% & 0.1\% & 19.0\% & 0.0\% & 6.8\% & 3.7\% & 0.0\% &
Finland & 107 & 20.2\% & 36.5\% & 24.1\% & 0.1\% & 0.0\% & 6.2\% & 3.0\% & 8.1\% & 0.0\% & 1
South Australia & 132 & 0.7\% & 0.0\% & 42.6\% & 33.7\% & 0.0\% & 0.0\% & 13.3\% & 9.0\% & 0
Spain & 132 & 17.1\% & 24.2\% & 25.1\% & 8.0\% & 0.0\% & 2.0\% & 18.8\% & 1.3\% & 0.2\% & 0.3
Belgium & 147 & 1.3\% & 39.6\% & 25.2\% & 3.6\% & 0.0\% & 2.8\% & 19.4\% & 1.7\% & 0.1\% & 4
Tasmania & 162 & 49.0\% & 0.0\% & 22.6\% & 10.8\% & 0.0\% & 0.0\% & 1.5\% & 16.1\% & 0.0\% &
East Denmark & 184 & 6.4\% & 5.5\% & 48.4\% & 1.3\% & 0.0\% & 16.8\% & 7.7\% & 10.8\% & 1.4\%
West Denmark & 188 & 8.8\% & 2.2\% & 56.3\% & 1.6\% & 0.0\% & 7.6\% & 8.5\% & 13.0\% & 0.9\%
Great Britain & 214 & 3.8\% & 12.4\% & 35.9\% & 2.7\% & 0.0\% & 6.2\% & 35.1\% & 2.0\% & 0.0
Netherlands & 218 & 1.1\% & 3.9\% & 46.7\% & 10.8\% & 0.0\% & 4.6\% & 22.4\% & 8.6\% & 0.8\%
New York ISO & 275 & 23.7\% & 22.8\% & 4.9\% & 0.0\% & 0.0\% & 0.1\% & 46.9\% & 0.0\% & 0.0\%
```

```
Italy (North) & 307 & 22.7\% & 14.5\% & 3.9\% & 2.9\% & 0.2\% & 3.1\% & 38.4\% & 1.5\% & 0.2
California & 328 & 8.4\% & 12.7\% & 7.9\% & 12.0\% & 3.0\% & 1.8\% & 48.5\% & 2.1\% & 0.0\% &
Germany & 389 & 4.4\% & 2.8\% & 39.7\% & 3.3\% & 0.0\% & 8.7\% & 14.4\% & 23.3\% & 0.6\% & 0
Ireland & 389 & 3.7\% & 0.8\% & 38.5\% & 0.2\% & 0.0\% & 2.5\% & 42.4\% & 9.7\% & 2.0\% & 0.
Western Australia & 417 & 0.0\% & 0.0\% & 14.1\% & 33.8\% & 0.0\% & 0.3\% & 24.2\% & 27.1\% &
Texas & 432 & 0.0\% & 9.1\% & 22.3\% & 6.0\% & 0.0\% & 0.0\% & 46.1\% & 16.1\% & 0.0\% & 0.4
Alberta & 447 & 1.9\% & 0.0\% & 12.4\% & 1.1\% & 0.0\% & 2.5\% & 70.7\% & 7.2\% & 0.0\% & 4.
Victoria & 508 & 3.9\% & 0.0\% & 17.5\% & 19.0\% & 0.0\% & 0.0\% & 0.3\% & 59.1\% & 0.0\% &
New South Wales & 578 & 3.2\% & 0.0\% & 9.5\% & 23.7\% & 0.0\% & 0.2\% & 0.7\% & 62.6\% & 0.
Queensland & 662 & 1.9\% & 0.0\% & 3.8\% & 21.1\% & 0.0\% & 0.0\% & 7.2\% & 65.7\% & 0.2\% &
South Africa & 685 & 2.2\% & 4.3\% & 5.8\% & 3.8\% & 0.0\% & 0.0\% & 0.0\% & 79.9\% & 2.0\% &
India (North) & 693 & 9.3\% & 2.2\% & 0.1\% & 10.6\% & 0.0\% & 0.0\% & 1.8\% & 75.2\% & 0.0\%
\bottomrule
\end{tabular*}

\end{table}
```

```
/opt/hostedtoolcache/Python/3.10.20/x64/lib/python3.10/site-packages/great_tables/_utils_rend
  warnings.warn(msg)
```

```python
import polars as pl
import polars.selectors as cs
from great_tables import GT, loc, style

coffee_sales = pl.read_ndjson("../examples/_data/coffee-sales.ndjson")

sel_rev = cs.starts_with("revenue")
sel_prof = cs.starts_with("profit")

# yo

print(
    GT(coffee_sales)
    .tab_header("Sales of Coffee Equipment")
    .tab_spanner(label="Revenue", columns=sel_rev)
    .tab_spanner(label="Profit", columns=sel_prof)
    .cols_label(
        revenue_dollars="Amount",
        profit_dollars="Amount",
        revenue_pct="Percent",
        profit_pct="Percent",
        monthly_sales="Monthly Sales",
```

```
        icon="",
        product="Product",
    )
    # formatting ----
    .fmt_number(
        columns=cs.ends_with("dollars"),
        compact=True,
        pattern="${x}",
        n_sigfig=3,
    )
    .fmt_percent(columns=cs.ends_with("pct"), decimals=0)
    # style ----
    .tab_style(
        style=style.fill(color="aliceblue"),
        locations=loc.body(columns=sel_rev),
    )
    .tab_style(
        style=style.fill(color="papayawhip"),
        locations=loc.body(columns=sel_prof),
    )
    .tab_style(
        style=style.text(weight="bold"),
        locations=loc.body(rows=pl.col("product") == "Total"),
    )
    # .fmt_nanoplot("monthly_sales", plot_type="bar")
    # .fmt_image("icon", path="docs/examples/_data/coffee-table-icons/")
    .sub_missing(missing_text="")
    .as_latex()
)
```

```
\begin{table}
\caption*{
{\large Sales of Coffee Equipment}
}

\fontsize{12.0pt}{14.4pt}\selectfont

\begin{tabular*}{\linewidth}{@{\extracolsep{\fill}}llrrrrc}
\toprule
 & \multicolumn{2}{c}{Revenue} & \multicolumn{2}{c}{Profit} &  \\
\cmidrule(lr){2-3} \cmidrule(lr){4-5}
 & Product & Amount & Percent & Amount & Percent & Monthly Sales \\
```

```
\midrule\addlinespace[2.5pt]
grinder.png & Grinder & \$904K & 3\% & \$568K & 4\% & [521, 494, 596, 613, 667, 748, 765, 686
moka-pot.png & Moka pot & \$2.05M & 7\% & \$181K & 1\% & [4726, 4741, 4791, 5506, 6156, 6619
cold-brew.png & Cold brew & \$289K & 1\% & \$242K & 2\% & [244, 249, 438, 981, 1774, 2699, 26
filter.png & Filter & \$404K & 1\% & \$70.0K & 0\% & [2067, 1809, 1836, 2123, 2252, 2631, 256
drip-machine.png & Drip machine & \$2.63M & 9\% & \$1.37M & 9\% & [2137, 1623, 1971, 2097, 25
aeropress.png & AeroPress & \$2.60M & 9\% & \$1.29M & 9\% & [6332, 5199, 6367, 7024, 7906, 87
pour-over.png & Pour over & \$846K & 3\% & \$365K & 2\% & [1562, 1291, 1511, 1687, 1940, 2177
french-press.png & French press & \$1.11M & 4\% & \$748K & 5\% & [3507, 2880, 3346, 3792, 390
cezve.png & Cezve & \$2.51M & 9\% & \$1.97M & 13\% & [12171, 11469, 11788, 13630, 15391, 1653
chemex.png & Chemex & \$3.14M & 11\% & \$818K & 6\% & [4938, 4167, 5235, 6000, 6358, 6768, 71
scale.png & Scale & \$3.80M & 13\% & \$2.91M & 20\% & [1542, 1566, 1681, 2028, 2425, 2549, 25
kettle.png & Kettle & \$756K & 3\% & \$618K & 4\% & [1139, 1023, 1087, 1131, 1414, 1478, 1456
espresso-machine.png & Espresso Machine & \$8.41M & 29\% & \$3.64M & 25\% & [686, 840, 618, 5
None & Total & \$29.4M & 100\% & \$14.8M & 100\% & None \\
\bottomrule
\end{tabular*}

\end{table}

/opt/hostedtoolcache/Python/3.10.20/x64/lib/python3.10/site-packages/great_tables/_utils_rend
  warnings.warn(msg)
```